

The Price of Proximity: Analysis on Land Rent and Accessibility Trends in San Francisco

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Section I: Introduction and Historical Development of San Francisco's CBD

1.1: Introduction to Bid-Rent Theory and the San Francisco Case

Understanding San Francisco's land-rent gradient, and its evolution, from its early industrial origins to the post COVID era requires us to analyze the interactions within the city between accessibility, economics, and work patterns. By applying bid-rent theory, we are able to gather and establish a foundation for our analysis of this gradient, letting us delve into how firms and households compete for central locations based on proximity value. According to the classic monocentric model, rents should decline with distance from the Central Business District (CBD), because being central reduces commuting cost, transportation costs, and access to potential clients and markets. As people move outwards, these advantages obviously diminish, creating a downward sloping rent gradient. San Francisco, and the Bay Area as a whole offer us a rather fresh perspective into this model. San Francisco is a peninsula encapsulated by steep hills and rather limited developable land, most recognizable for its transition from port and manufacturing to finance, and more recently, information-technology industries. These factors have all played into creating one of the steepest rent gradients in the United States, trademarked by its high downtown office values, residential prices, and housing scarcity. Post pandemic, the city shifted towards remote and hybrid work, further complicating this pattern, furthering the argument that classical bid-rent curves are restructuring. By examining and looking into these historical and now modern changes, we can gain some type of insight into how the previously mentioned factors are redefining and changing San Francisco's urban structure.

1.2: Historical Origins of San Francisco's Urban Form (1848–1870s)

San Francisco's urban form became most prominent during the 1848-1855 Gold Rush, when a small mining settlement was turned into a commercial gateway for the Pacific mining frontier. Characterized by a deep natural harbor, and good placement along the Bay, the waterfront became a critical point for early economic development.

Soon enough, the shoreline turned into a dense, compact location filled with warehouses, shipping offices, banks, and suppliers, financed by arriving ships all anchored along the Embarcadero, adjacent to the old time CBD. This concentration works in perfect accord with the bid-rent dynamic. Land closest to the port allowed for access to trade, labor and goods movement, all highly sought after by booming businesses. Firms whose profit was dependent on minimized transport times and costs were far more willing to outbid others for these sites. Because of the scarcity of waterfront land, and the essential role it could play for more commercial activity, these parcels created some of the steepest and most recognizable land-rent curves in America.

1.3: Formation of the Original CBD (1870s–1900)

By the later 19th century, San Francisco really honed in on becoming a port-industrial economy. The waterfront, the initial economic core of the city, hosted a wide variety of businesses, ranging from shipping and cargo, to fish processing and food packing, even weaning into some of the earlier financial institutions which were tied to trade. Due to the transportation dependent nature of these businesses, the waterfront was reinforced as the original CBD, as firms paid premium rents. This waterfront had become the industrial center, the commercial core, and the Bay Area's major labor market junction.

In order to accommodate San Francisco's growing role as a major market, planners created a more rigid street grid, which seemed to clash harshly with the peninsula's incredibly challenging topography, made up of a plethora of hills. The physical constraints limited development to the areas of flat low land near the water. Because this small zone had the most accessible and buildable land, competition only intensified. The geography raised the cost of moving outward from the port, as firms and households that located farther had to face steeper topography, and lower quality land, furthermore emphasizing the importance of proximity within SF. This terrain also shaped urban growth in the Bay, as industrial users would dominate the waterfront and flat areas nearby, leaving the residential development into the hills. This led San Francisco

to develop into the classic monocentric city model, leaving the CBD to be tied to the highest land value, while rent declines rather predictably with distance outward.

Transit innovations further reinforced the dominance of the CBD. The introduction of streetcars and cable railways extended the feasible commuting radius for workers. Through its introduction, households were able to travel for a lower cost of distance, but time-sensitive firms still held that premium price on centrality, further strengthening San Francisco's monocentric layout. These innovations allowed residential neighborhoods to expand outwards, further away from the valuable parcels of land, greatly intensifying commercial land use near the waterfront, and especially the Financial District, a dense cluster of high-value commercial activity formed adjacent to the Embarcadero and Ferry Building. This area exemplified the structural shift towards office based commerce, and not sole reliance on the port to be the entire CBD anymore. Multi-story construction became prominent within this Financial District, as firms were more willing to substitute land for capital in response to higher land prices. Commercial activities with higher time sensitivity due to required face to face time clustered towards the center, while households, much less reliant on proximity, were moved outwards to other neighborhoods. This, paired with the vertical development in high bid price areas were exhibitions of the most crucial and defining traits of monocentric urban form within the Bay Area.

1.4: Transformation of Economic Base and the Expanding CBD (1945–2000s)

After World War II, San Francisco's economic landscape was changed, highlighting differences made to the CBD structure and rent within the area. Manufacturing and major port activity moved to larger facilities and buildings across the Bay, with a major area of move being to Oakland. This led to a decrease in waterfront demand, reducing transportation dependent activity, which had created that competitive advantage for the area. Although this location's land values were defined by historical maritime importance, the rent gradient did not flatten. Instead, opportunity arose for new types of accessibility reliant users such as finance, insurance, real estate, and corporate services. These operations all are heavily dependent on face to face interaction, rapid

information exchange, and centrality, reinforcing the demand within the Financial District.

In the late 1990s, the tech boom that swept the Bay Area furthered the transformation of the CBD and surrounding areas, especially SoMa, a vast warehouse filled district adjacent to the financial district. Tech firms and startups didn't necessarily need adjacency to the financial district, but they placed a massive emphasis on connectivity, concentration of talent, and flexible office space, which led to their takeover of the area. The firms began bidding for former industrial buildings surrounding the previous CBD, thus producing a secondary high-rent employment cluster right next to the traditional CBD. This shift created more of a polycentric city model, clashing with our classic bid-rent model. Although the Financial District was still the highest rent commercial zone, SoMa had rapid employment growth, altering accessibility premiums across a broader area. Land values continue to rise sharply within both districts, no longer driven by maritime access or manufacturing needs that had spurred the original importance of the area, but instead by proximity to labor markets, transit centers, reliant on BART and Muni Metro, as well as agglomeration sectors.

These developments showed how San Francisco's CBD had completely reoriented its economic core. The transition from industrial reliance on ports, to finance, and then technology reshaped accessibility, employment, and land rent within the area. The city did remain mostly monocentric, though the structure of the CBD itself had expanded and most definitely diversified to several different sectors, allowing for multiple nearby locations with high "willingness-to-pay". This created the baseline upon which San Francisco would continue to blossom even further, characterizing its urban core through its steep rent gradients until present day. This paper seeks to understand whether these modern urban and commercial land rent trends align with bid rent theory discussed in class, as well as determine how these patterns have shaped the central business district since the COVID-19 pandemic.

Section II: Residential Bid Rent Theory Applied to Modern San Francisco County

Having established a background on bid rent theory and its applications towards land rent prices in monocentric metropolitan areas, we must now analyze the validity of said theories in the context of modern San Francisco (SF) county. It is worth noting that all analyses done in this section applies only directly to neighborhoods located within the official boundaries of SF county and disregards all other population centers within the Bay Area. This is done primarily because of the presence of both the Pacific Ocean and the San Francisco Bay surrounding the county on three sides. Any bid rent analysis using downtown SF as the central business district extended towards other areas of the bay would be inaccurate due to the presence of these bodies of water and the existence of other large population and business agglomerations in these areas. SF County measures approximately 12km by 12km (7 square miles) and is of significant enough size to perform bid rent analysis. Other than the surrounding coastline as geographical barriers, the San Mateo County line is used as the southern border for the following analysis for both simplicity and convenience when working data.

2.1: Population Density and Locational CBD of San Francisco County

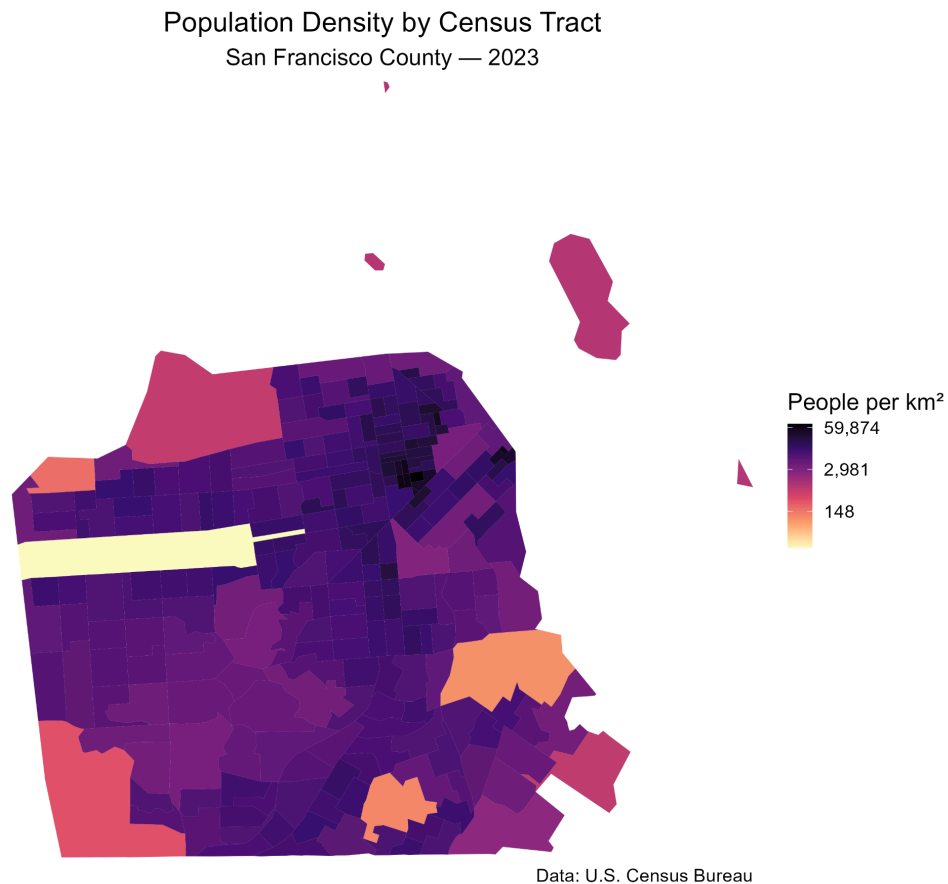
Section I previously identified the Financial District of downtown SF present in the northeast corner of the county as the current modern central business district for the area. The emergence of the technological and financial sectors as the primary industries for SF shifted the highest agglomeration of firms to this area from its previous center in the port area of Embarcadero. Bid rent theory assumes that the density of not only firms but people increases uniformly with proximity to the city's central business hub. In order to determine the validity of any further bid rent analysis on the area, it

must be determined as to whether or not the population of SF county is dispersed as such.

The United States Census Bureau annually conducts a varied and comprehensive national survey across the nation, collecting a variety of data on population centers such as density, median household income, and many other relevant demographics. While much of their data is cured and the household level, many variables are categorized into census tracts before being publicly available for release. Census tracts are designated geographical blocks in which census data is organized into each year so that it may be properly analyzed. Tracts are not determined based on neighborhood, political, or zip code distinctions but rather population; each tract is supposed to contain roughly 2000-4000 people and are held constant over time in order to allow for long term economic and demographic analysis.

San Francisco County is a relatively large density area compared to other parts of the nation, and therefore is made up of 196 individual small tracts. The American Community Survey (ACS) compiles publicly available census data into 1 and 5 year summaries. For this figure and all subsequent visuals in this section, we used a statistical programming language called R to create visual representations of the census tract data, which is only available in CSV exports and is incredibly difficult to analyze by hand. Census tract data sets are divided by individual variables (such as population), but contain data for every single region of the United States. By conducting our analysis in R, we are able to easily specify the data for just San Francisco County and develop easily understandable visuals. In order to understand the population distribution of the area, we drew upon ACS's most recent data (2023) concerning census tract population and size. Using these two variables, we calculated the population density per tract and imposed it on a tract map of the county.

Figure 1:



Though census tracts do not directly coincide with specific areas or neighborhoods in the county, it is immediately apparent that the population center of the county coincides with the relative location of the financial district in the downtown area. The densities in that area are the darkest relative to any other part of the map, with multiple census tracts in vicinity reaching over 10,000 people/km². Upon initial inspection of Figure 1, the rest of the county does seem to disperse in population density as you move west and south from the financial district, with mid distance areas such as Mission, Castro and Haight Ashbury having relatively higher densities than

outskirts neighborhoods such as Twin Peaks, Oceanview, and Parkside. In order to properly determine if population density is positively associated with proximity to the financial district, we developed a distance from financial district versus population density scatterplot that plotted each tract within the county as a data point.

Figure 2:

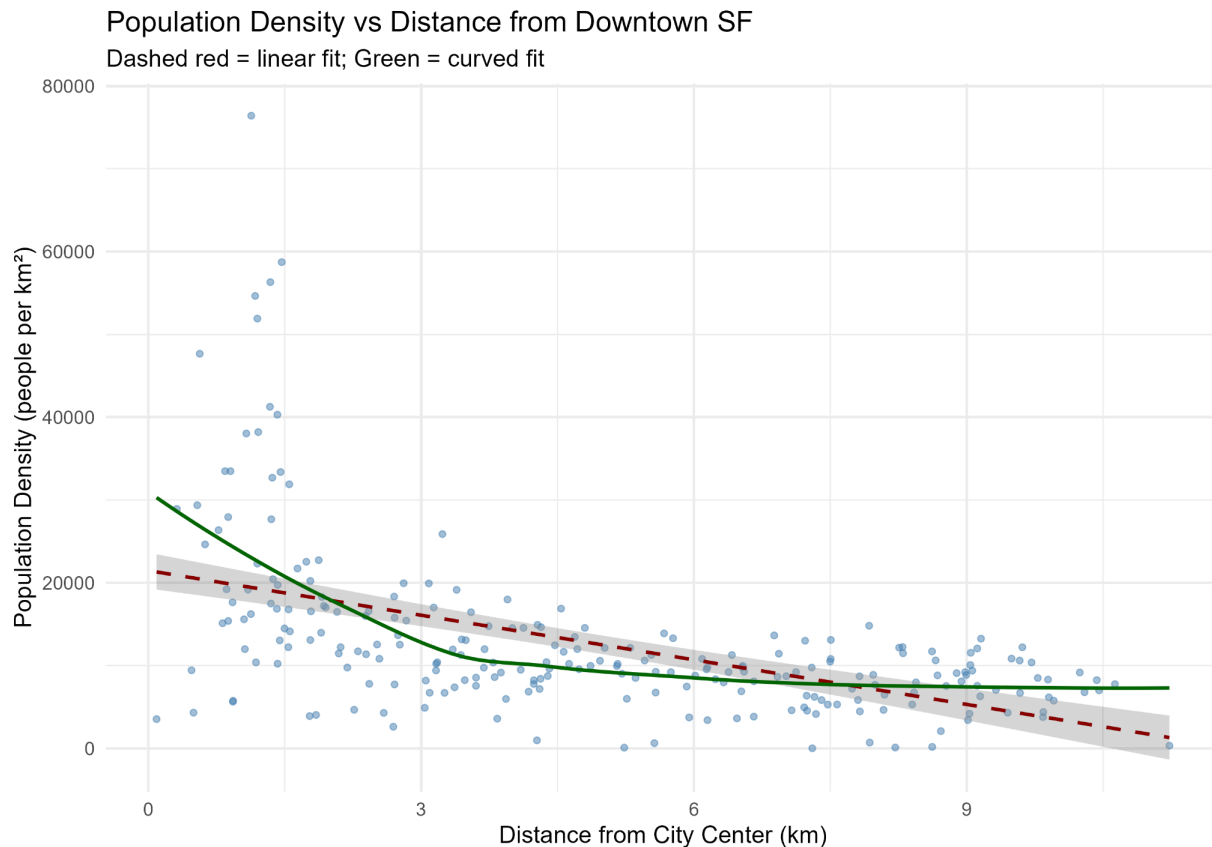


Figure 2 places the specific location of the CBD at the corner of Market Street and Pine Street, due to its proximity to firm dense buildings such as Salesforce Park and it being generally located in the center of the financial district. Both best fit lines indicate that a strong positive relationship between density and distance exists, with a particularly large increase in density for many tracts within the first 1.5 kilometers from the city center. Pulling from the same data used to construct Figure 2, we can determine that SF County has a density gradient of approximately 0.205. This means that for every mile away from the financial district, the population density decreases by roughly 20%. This is higher than the typical range for major U.S. metropolitan areas (range 0.05 to

0.15 according to Chapter 7 Slides) suggesting a particularly significant level of clustering around the financial district. Ultimately, our analysis of the population distribution of the county reinforces the claim that the financial district is a suitable location for San Francisco County's monocentric CBD and suggests that the population is distributed appropriately for bid rent analysis.

2.2: Residential Bid Rent Analysis For San Francisco County

Now that we have determined that the demographic and geographic conditions of San Francisco County are suitable for bid rent analysis, we will conduct a graphical analysis of residential rent prices in the county to determine if they align with trends proposed by the bid rent theory presented in our class. As previously mentioned, residential bid rent theory proposes that household rent per a given unit of space (i.e. hectares, square feet) increases with proximity to a city's CBD as residents save on commuting costs. ACS collects census data on the median household rent for each census tract in the county which, similarly to Figure 2, can be visualized by plotting each census tract's median rent in reference to its distance from the financial district.

Figure 3:

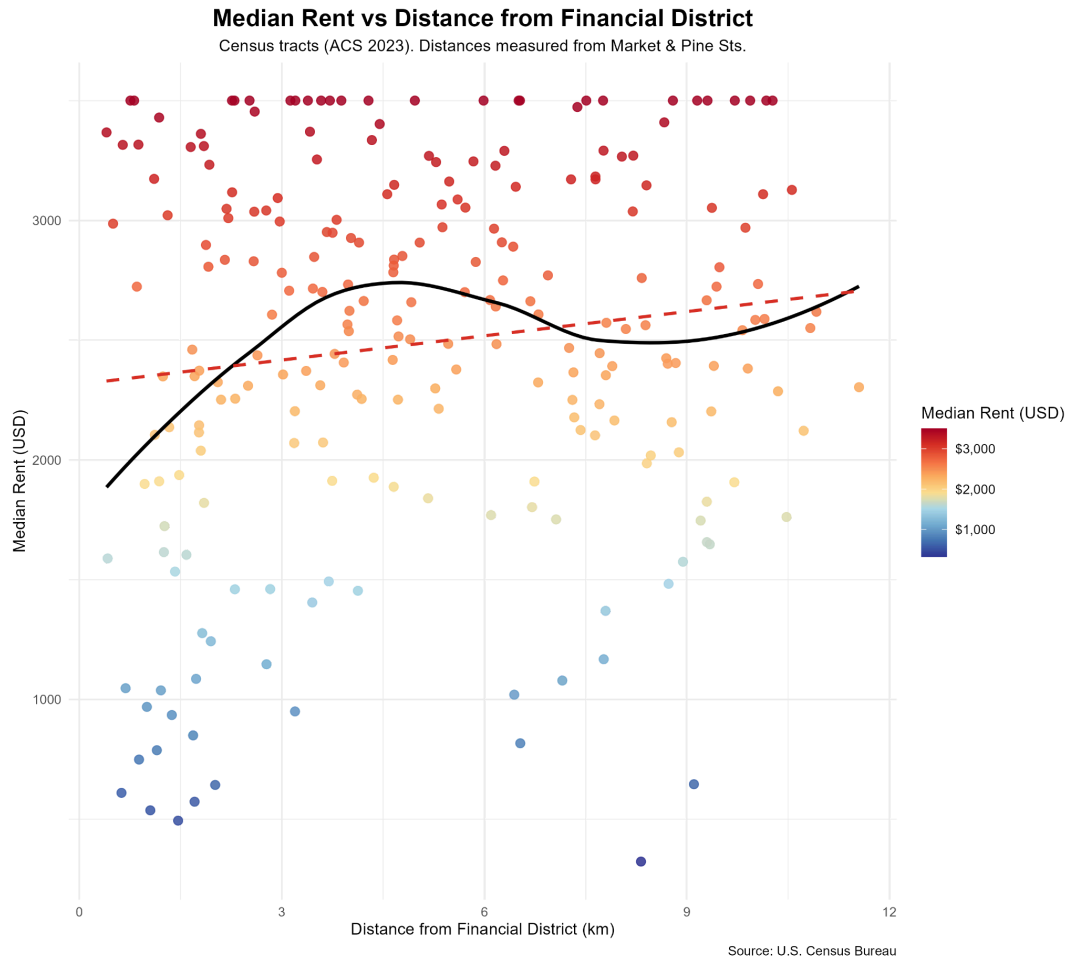
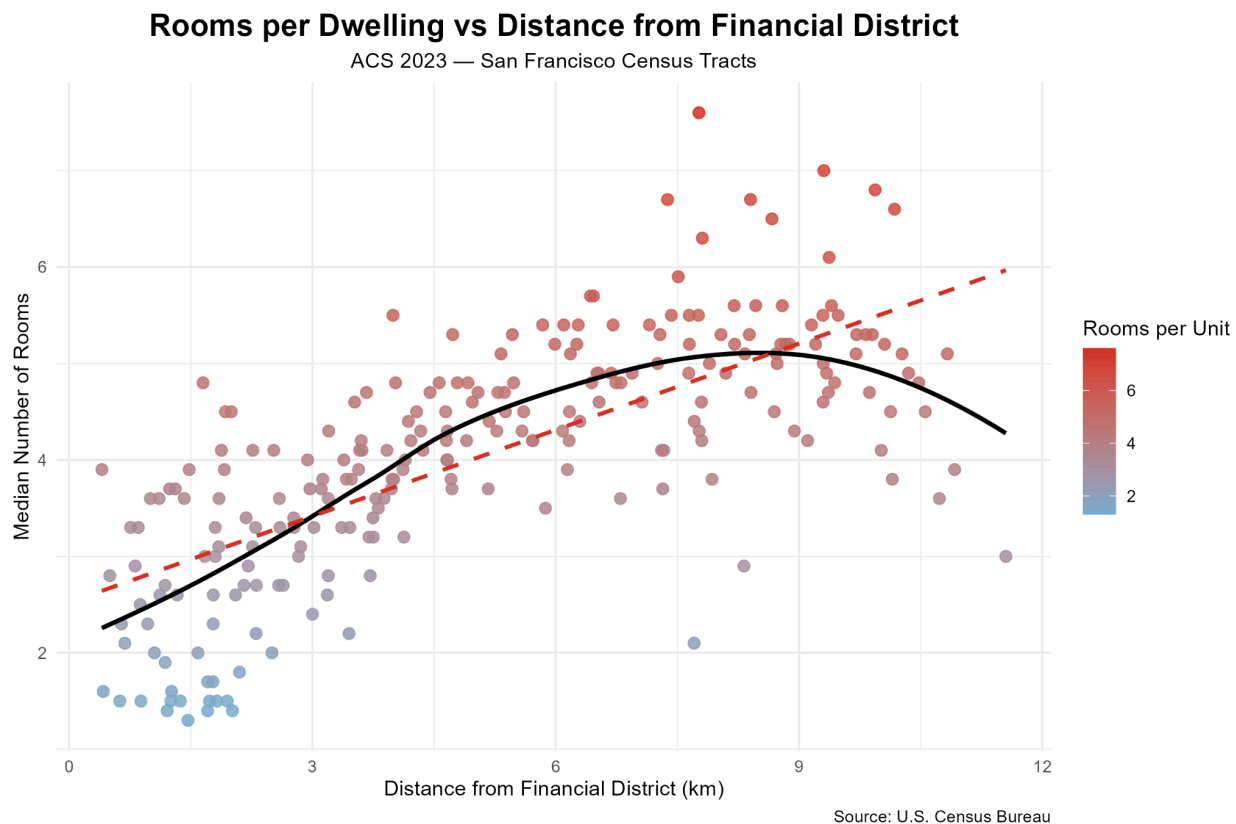


Figure 3 suggests that overall median household rent increases slightly with distance from the CBD. The exact equation of the linear regression line has median rent at \$2316 at the CBD, with rent increasing at approximately \$34 per kilometer from the CBD. It is worth mentioning that the strength of the linear regression model is exceedingly weak; with an r -squared value of 0.017 (1 is the strongest measure), data from the graph is not clustered closely to the regression line. Multiple tracts within 3km of the CBD have median rents of +\$3000 dollars and a few outlier tracts +6km from the CBD have rents below \$1000.

Though the existence of typically higher trending rents further from the CBD initially seems contradictory to established bid rent theory, this graph does not account for consumer substitution. Consumer substitution dictates that residents who locate further from the CBD capitalize on lower land costs by purchasing/renting larger lot sizes than residents closer to the center. This gives rise to the concept of the suburbs, where typically higher income individuals (usually families) locate themselves outside the center in more spacious dwellings than lower income (usually single occupant) households. This trend explains the presence of higher rents in tracts located further from the financial district, where households are presumably consuming more land per dwelling, therefore incurring larger rents overall. In order to develop an accurate bid rent representation of SF county, median rent needs to be calculated in terms of a consistent unit of space so that higher land costs close to the center can be properly visualized.

The ACS publicly available census data does not contain any variables regarding house square footage or lot size for dwellings. This makes it difficult to contextualize rents in different parts of the county to each other as there is no available information on the difference in dwelling size. Other public datasets such as SF Building Footprints (used later for commercial analysis in section 3) contain data on lot size, but do not distinguish between types of land use. Using such data would result in a flawed analysis as large commercial buildings or apartment blocks near the CBD would distort the presumably smaller typical dwelling size. However, the ACS data does contain a metric for the median number of rooms per dwelling in each census tract. Plotting each of these tracts in reference to their distance from the financial district gives us a preliminary insight into the behavior of this variable.

Figure 4:

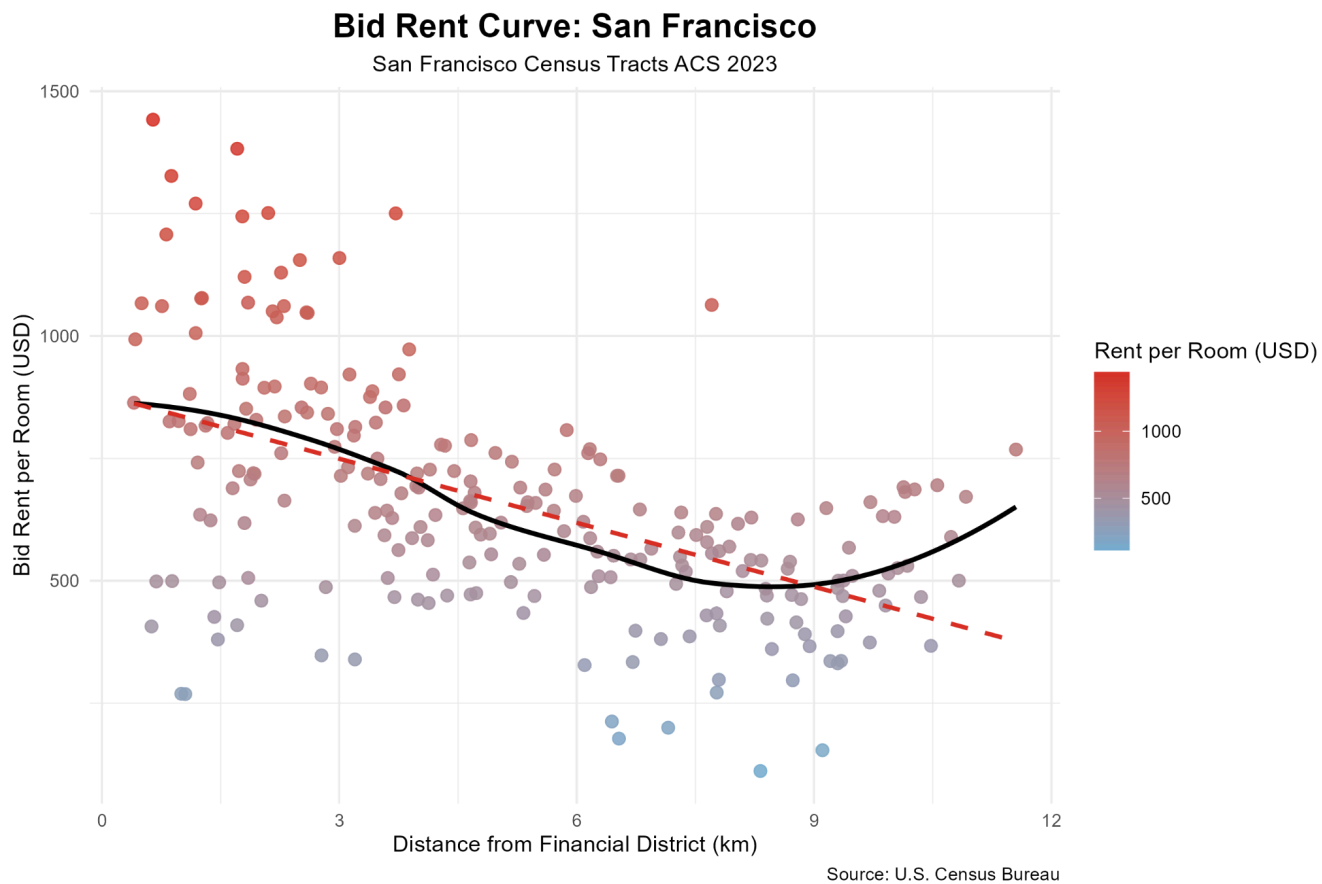


It is immediately evident from the plot that there is a strong positive correlation between rooms per dwelling and distance from the CBD. The r-squared value of the plot is 0.515, indicating a relatively solid relationship between the two variables (values above 0.7 are rarely seen in social science). The presence of very few outliers in the graph also reinforces the validity of its findings; no census tracts within 3km of the CBD have a median of 5 rooms or larger, and only two tracts beyond 6km have medians lower than 3 rooms. The linear regression line places the typical number of rooms for a residence inside the CBD (0km) as 2.5, increasing by approximately 0.3 rooms for every kilometer away from the financial district. This suggests that typical dwellings 10km from the center will have approximately 5.5 rooms.

Figure 4 presents a clear argument for the widespread presence of consumer substitution trends in SF County as it links increased distance from the financial district with increased dwelling size. The graph also suggests that using a singular room as the spatial unit to approximate lot size when calculating bid rent could be viable. Though it is obviously not perfect (not all rooms are of equal size), the lack of other publicly available data necessitates its use. If median tract rent is treated as the approximate

willingness to pay residents for a plot of land in a given tract, then, by combining the census variables used to create Figures 3 and 4, we are able to estimate the bid rent per room for each census tract in the SF County area. Essentially, data is pulled from the ACS dataset on each tract's median residential rent and divided by the median number of rooms for that tract. The resulting value, the bid rent per room of that given tract is plotted in reference to its distance from the financial district alongside all other tracts. The resulting regression lines from plotting all tracts in the county will make up an approximate bid rent curve for the SF County area.

Figure 5:



The equation for the linear regression line of Figure 5 is as follows:

$$\text{Rent per room} = \$879.877 - \$43.582 * \text{Distance from CBD (km)}.$$

Upon initial inspection, this seems to align with proposed bid rent theory. The price of residential land is highest near the financial district where commuting costs are presumably near zero. A majority

of the county's population locate themselves within 2 km of the center (see Figures 1 and 2) and land prices there are proportionally higher due to high demand. As land consumers in SF county locate further and further from the center, they garner higher commuting costs but save on lower land prices. Consumers in the county tend to choose larger lot sizes at these distances (see Figure 4) as the usually more affluent suburban homes prefer and afford the increased living space. There is an evident and direct tradeoff between price and proximity.

Despite the linear regression line aligning with typical monocentric bid rent trends, the black curved best line in Figure 5 highlights inconsistencies within the data. Typical bid rent graphs that account for consumer and factor substitution are convex, with exponentially higher levels of land rent close to the center and increasingly smaller drops in land value the further you travel from the CBD. These conditions do not hold at either of the extremes in the graph. As you approach the CBD in Figure 5, land prices become concave and level off, as a significant number of tracts still have rent prices per room lower than \$750. While the graph is appropriately convex 4-8km from the financial district, land rent prices trend upward from 8-12km from the financial district. Bid rent for land should be worth the least in the furthest tracts from the financial district and the most in the tracts closest to the center, but that simply isn't the case.

2.3: Reasons for Variations from Bid Rent Theory in San Francisco County

While the bid rent curve for SF county proposed in Figure 5 does not follow traditional models used in class, there are several reasons that explain the discrepancies at either end of the bid rent curve. For starters, variations in land rent in mid and far distance tracts can be explained by the abundance of coastal land in these areas. A majority of the census tracts greater than 9km from the financial district are either bordering or near the Pacific Ocean, which drives up the desirability for land in these areas, thus artificially increasing the prices of land further from the CBD. Standard bid rent models used in class assume a CBD located at the center of a city, which sprawls out uniformly in all directions until land use switches to agricultural land beyond

the city borders. This is simply not the case for San Francisco, which may partially account for the increase in land prices beyond 9km from the center.

Moreover, residential areas in and around the financial district suffer from negative externalities that lower the desirability of living, and by extension the residential land prices, in those areas. The San Francisco Planning Department data shows significantly lower concentrations of retail stores, parks, and schools, in the area than other parts of the county (SF Planning Department, 2023). Safety is also a major concern for residents in that area, as the SF Controller's Office reports the highest concentration of open-air drug use, street disorder, and homelessness within half a mile of the financial district's boundaries (SF Controller's Office, 2023). These drawbacks, along with other common disamenities from locating close to the CBD such as congestion and pollution, ultimately flatten the bid rent curve within 3km of the financial district.

2.4: Methodology Concerns

Certain flaws in the methods used to visualize and analyze the bid rent patterns in SF County may have also contributed to the discrepancies from standard city bid rent models. As stated previously, the use of room size as a substitute for a precise lot size measurement may have influenced the resulting curve, particularly in areas further from the CBD. It is likely that the standard room sizes in suburban dwellings further from the CBD are larger than those near the center, which may explain some of the higher bid rent per room medians further from the CBD. Additionally, using room count to measure dwelling size is inherently flawed due to the lack of accounting for open space such as backyards, lawns, and pools. These amenities are certainly more common in the suburbs far away from the city center, and all serve to increase lot size and rent price of homes in these areas, which artificially drives up the bid rent per room. Furthermore, the shape of the bid rent curve is determined by weighing each data point, and therefore census tract, equally. Since tracts are determined by population, there is a larger concentration of census tracts closer to the CBD, which skews the graph to place more weight on the land rent in areas closer to the center.

Section III: Commercial Real Estate, Commuting, and Zoning Trends of San Francisco County

In the modern day, the Central Business District (CBD) of San Francisco is found within the financial district and SoMa (South of Market) neighborhoods. Bordered on the west by the Embarcadero and Ferry Building, on the north by Washington Street and the famous Transamerica pyramid, and on the east and southeast by Chinatown, Union Square, and the newly constructed Salesforce Park. This area has become the creme de la creme of commercial real estate in the city, representing the majority of prominent F.I.R.E (Finance, Insurance, and Real Estate) firms, as well as many other large corporations and tech giants.

3.1: Transportation Infrastructure

The development of the aforementioned area as San Francisco's most desirable real estate for office firms is largely due to the transportation infrastructure, which makes this area accessible for bay area office workers. Its proximity to the Ferry building, interstate 80, and BART (Bay Area Rapid Transit) lines, allows workers to commute in from North, South, and East Bay, as well as from the rest of the city.

The ferry building, seen below in Figure 6, represents not only a 127 year old San Francisco landmark, but a key commuting hub shuttling workers from the North and East Bay right into the CBD. Located on the East side of the financial district, the Ferry Building is a terminal for ferry rides from two large operators, Golden Gate Ferry, and San Francisco Bay Ferry. The former, serves North Bay workers in Marin County, with ferries coming in from Larkspur, Tiburon, Sausalito, and Angel Island. Golden Gate routes ferried 1.4 million people in 2024, with average weekday ridership of 4,000 per day (Golden Gate Ferry, 2025). The San Francisco Bay Ferry routes carry East Bay commuters from stops in Alameda, Oakland, Harbor Bay, Richmond, and Vallejo, bringing across 7,148 passengers a day in 2024 and 2.6 million throughout the year (San Francisco Bay Ferry, 2025). The CBD of San Francisco is facilitated by its location directly adjacent to water, where office workers from North and East can take large

ferries right into the financial district and walk to their offices, bypassing traffic congestion.

Figure 6:



Source: (GoldenGate.org)

The ferry building is a key hub for commuters from across the bay into the heart of the San Francisco business district. The Interstate 80 and BART trainlines are an even more integral component of the CBD's commuting infrastructure. Interstate 80 (i80) is a highway that stretches across the country from East-West, terminating in San Francisco when it merges with US-101. This highway runs along the Bay Bridge

between Oakland and San Francisco, and is the primary way that workers from the East Bay commute into the city. It's conveniently located just a few minutes south of the financial district, and "is the region's workhorse bridge, carrying more than a third of the traffic of all of the state-owned bridges combined" (Metropolitan Transportation Commission, 2025). Interstate 80 diverges south from the bridge to unite with US-101, a massive highway extending south down the peninsula. Taking US-101 North to i80 is a straightaway shot for office workers down the peninsula to get right into the CBD. This all being said, it's easy to understand how these freeways become highly congested and parking in the CBD becomes very scarce. These issues cause many commuters to elect to take BART trains. BART is the primary rail/subway system in San Francisco, seen below in Figure 7, and has lines running from the peninsula, through the city, and deep into the East Bay. The Embarcadero and Montgomery BART stops are in the heart of the financial district, and each stop services 4 lines bringing passengers from as far down the peninsula as Millbrae, and all throughout the East Bay. Being in the CBD, these stops are the busiest in the BART network, with 20,192 and 16,514 commuters respectively on an average weekday this past October. (BART Monthly Ridership Snapshot, 2025).

Figure 7 & 8:



Ridership by Station (Average Weekday)					
Richmond	2,696	-36	-1.3%	+240	+9.8%
El Cerrito del Norte	4,347	-32	-0.7%	+315	+7.8%
El Cerrito Plaza	2,515	+72	+2.9%	+329	+15.0%
North Berkeley	2,037	+27	+1.4%	+217	+11.9%
Downtown Berkeley	7,260	+194	+2.8%	+1,088	+17.6%
Ashby	2,317	+15	+0.7%	+240	+11.5%
Antioch	2,041	+26	+1.3%	+10	+0.5%
Pittsburg Center	581	+9	+1.6%	+47	+8.8%
Pittsburg / Bay Point	2,594	-14	-0.5%	+311	+13.6%
North Concord / Martinez	698	-1	-0.2%	+35	+5.3%
Concord	2,969	-11	-0.4%	+164	+5.8%
Pleasant Hill	2,892	+8	+0.3%	+239	+9.0%
Walnut Creek	3,270	+21	+0.6%	+240	+7.9%
Lafayette	1,860	+24	+1.3%	+154	+9.0%
Orinda	1,478	-13	-0.9%	+158	+11.9%
Rockridge	3,192	+123	+4.0%	+450	+16.4%
MacArthur	4,442	+126	+2.9%	+518	+13.2%
19th St / Oakland	5,577	+165	+3.1%	+686	+14.0%
12th St / Oakland City Center	5,847	-67	-1.1%	+604	+11.5%
Berryessa / North San Jose	1,940	+31	+1.6%	+200	+11.5%
Milpitas	1,453	+10	+0.7%	+116	+8.6%
Warm Springs / South Fremont	1,356	-11	-0.8%	+127	+10.3%
Fremont	2,384	-8	-0.3%	+179	+8.1%
Union City	2,176	-16	-0.7%	+176	+8.8%
South Hayward	1,617	+13	+0.8%	+135	+9.1%
Hayward	2,525	-45	-1.8%	+33	+1.3%
Dublin / Pleasanton	3,434	-46	-1.3%	+277	+8.8%
West Dublin / Pleasanton	1,432	-12	-0.9%	+84	+6.2%
Castro Valley	1,419	+13	+0.9%	+135	+10.5%
Bay Fair	2,903	+3	+0.1%	+139	+5.0%
San Leandro	3,728	+1	+0.0%	+168	+4.7%
Coliseum	2,779	-106	-3.7%	+176	+6.7%
Oakland International Airport	558	-21	-3.6%	-81	-12.6%
Fruitvale	4,364	-9	-0.2%	+34	+0.8%
Lake Merritt	3,205	-3	-0.1%	+143	+4.7%
West Oakland	4,215	+134	+3.3%	+609	+16.9%
Embarcadero	20,192	-44	-0.2%	+1,924	+10.5%
Montgomery Street	16,514	+419	+2.6%	+1,923	+13.2%
Powell Street	12,785	+838	+7.0%	+1,357	+11.9%
Civic Center	10,891	+650	+6.3%	+1,286	+13.4%
16th Street Mission	6,533	+430	+7.0%	+684	+11.7%
24th Street Mission	6,157	+268	+4.5%	+322	+5.5%
Glen Park	3,527	+93	+2.7%	+175	+5.2%
Balboa Park	4,711	-25	-0.5%	+315	+7.2%
Daly City	5,832	+10	+0.2%	+298	+5.4%
Colma	1,764	+63	+3.7%	+185	+11.7%
South San Francisco	1,596	+28	+1.8%	+141	+9.7%
San Bruno	1,724	+50	+3.0%	+157	+10.0%
San Francisco International Airport	4,662	+415	+9.8%	+571	+14.0%
Millbrae	2,164	+52	+2.5%	+288	+15.3%

Source (Bart.gov)

3.2: Commercial Real Estate Landscape

All of this commuter infrastructure makes the location of San Francisco's CBD ideal for office firms. Through the ferries, highways, and BART trains, this central location reduces overall commuting times for office workers from in the city and around the Bay. Lower commuting time almost always means lower commuting cost, and lower commuting cost for workers means lower wages for firms, when employers' wages need to account for lower employee travel cost. The large pool of educated and high-skilled workers in the Bay Area, made accessible by downtown's transportation infrastructure, along with the agglomeration of all the prominent office firms already in the CBD, make

land rents in the CBD exorbitant. In fact, Commercial Real Estate firm Cushman & Wakefield reported Class A rents in the CBD reached \$73.98 per square foot for Q3 of 2025 (Cushman & Wakefield 2025). Class A rents refer to modern buildings with parking, HVAC, and other amenities. These comprise the real estate that top firms build their offices in. Being San Francisco, average class A rents outside the CBD were still high, but noticeably lower at \$67.05 per square foot. Class A rents in the CBD can potentially exceed the average by a substantial amount for desirable buildings in large transactions. Recently, North Financial District buildings on 420 and 505 Montgomery Street sold to two different real estate development firms at \$134 and \$299 per square foot respectively (Cushman & Wakefield, 2025).

VC-backed AI companies have been another huge driver of commercial real estate prices in the San Francisco CBD. Needing to expand quickly and having unpredictable future-growth outcomes, startups almost always lease, or even sublease office space. VC funding in the first three quarters of 2025 reached almost \$100 billion (Cushman & Wakefield, 2025), almost all of which towards AI startups, leaving these new firms with piles of cash to rent out the best office space. They have certainly done that, leasing “more than 5 million sq. ft. of San Francisco office space over the past five years and are projected to take up... 2.7 million annually through 2030” (CBRE, 2025). This trend is supported by VC-backed tech firms of the dot-com bubble and 2010s, which took on larger office leases in order to scale business capacities. Recently, four of the largest eight leasing transactions in San Francisco in Q3 of 2025 involved venture backed AI companies. These transactions represented around 280,000 square feet of real estate in the South Financial District and Mid-Market areas, including new leases, sub-leases, and renewals (Cushman & Wakefield, 2025).

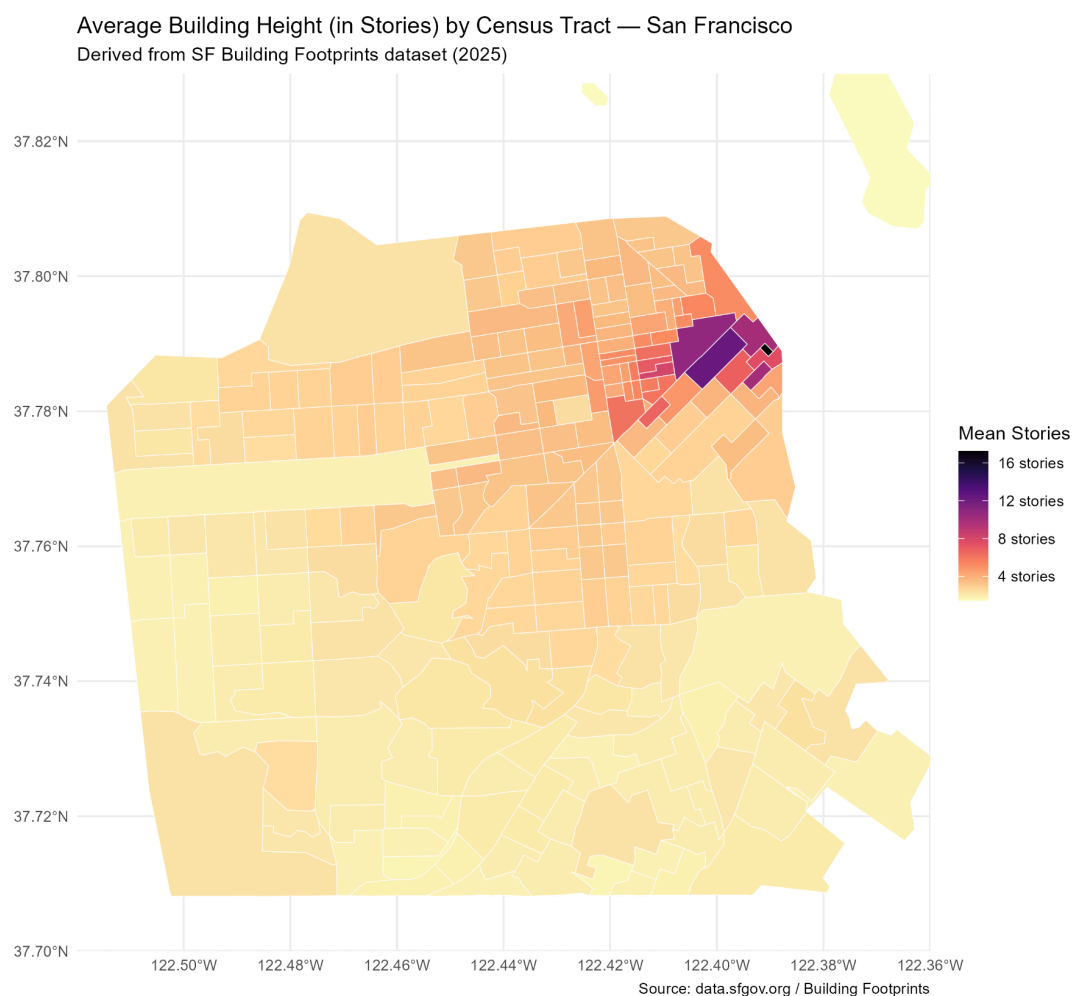
3.3: Building Heights and Zoning

When a downtown area becomes highly-demanded by office firms, and CBD land-rents soar, vertical construction becomes the only solution to attempt to accommodate the quantity of demanding firms. This concept is known as factor substitution, where the land-owners or developers substitute the use of land for capital.

Land is very expensive in the CBD of San Francisco, but the cost of building towering office buildings is relatively fixed. Erecting sky-scraping office buildings allows land owners to offer firms a large square footage of office space, while only owning a small plot of expensive land. Factor substitution allows for more office firms to agglomerate in the CBD, as it makes per square foot office rent cheaper than if all buildings were to have only 5 floors for example. However, it makes the land-rent of plots of land in the CBD extraordinarily high since they can accommodate a higher number of rent-paying tenants.

Figure 9 below displays the average building height of census tract areas within San Francisco. The vast majority of the city is shaded in light or medium tan, corresponding to buildings 1-5 stories high on average. The black, purple, and pink sections in this graph represent tracts of the CBD such as the North and South Financial District, SoMa, and Union Square, where the average height is between 10-16 stories. On these sought after plots of land, office firms stack up dozens of stories high in downtown skyscrapers to take advantage of the accessible high-skilled labor pool and close contact with peer firms.

Figure 9:



The knowledge and labor pool sharing that occurs between companies in the same industry can drive massive value. This phenomenon is what causes office firms to so desperately desire to rent space in the CBD. In San Francisco, businesses choose not only to locate near each other in the CBD, but even further in the same building. A quintessential example of this is the 555 California Street building in the Financial District. Formerly known as Bank of America Center, this 52-story tower was the tallest building west of the Mississippi upon its construction in 1969 (The Trump Organization, 2025). Its tenants now include top law firms Kirkland & Ellis, Jones Day, and Sidley Austin, as well as premier banks Morgan Stanley, UBS, and Bank of America and consulting giant, McKinsey & Company (Vornado Realty Trust, 2025). The proximity of these companies is mutually beneficial as they can all benefit from joint and

referral business. For instance, say Coca Cola acquires another large beverage company, Morgan Stanley's investment bank can handle the acquisition financing, Kirkland & Ellis can serve as legal counsel, and afterwards McKinsey can provide restructuring or integrative consulting services. There is no better way to have face-to-face contact and information sharing with other firms than sharing an elevator. Buildings like 555 California and other office hubs, as well as tall apartment rises for office workers who desire a commute by foot constitute the iconic San Francisco skyline found in the CBD.

While inner-city skyscrapers attract business, drive corporate productivity, and produce an attractive skyline, there are also negative externalities associated with tall building heights. Tall buildings house or employ large numbers of people relative to their lot-size, exploding demand for car parking, leading to a scarcity of spots. Furthermore, tall buildings block sunlight from people on the streets and in lower housing, as well as blocking views from adjacent buildings. For these reasons, many San Francisco residents elect to live in the east and northern parts of the city, away from downtown. San Francisco's most expensive neighborhoods, Pacific Heights, Presidio Heights, and Sea Cliff all reinforce this theory. These neighborhoods and others in San Francisco are able to restrict building heights, commercial development, and even lot-size through zoning codes. The San Francisco Planning Department is tasked with apportioning every block of San Francisco for different uses and regulating building heights and size within different zones. Below are official maps from the San Francisco Planning Department, with Figure 10 representing zoning use districts and Figure 11 representing building height/bulk districts. In Figure 10, the red blocks represent zoning for office space and retail, with the surrounding orange blocks zoned for downtown residential housing. Those same blocks in Figure 11 are colored brown and red, corresponding to the highest building limits in the city. The San Francisco Planning Department has zoned the CBD to allow for the largest buildings and designated office and commercial areas. Moving away from the CBD in Figure 10, we see mostly yellow shaded blocks, constituting residential zoning, with veins of pink blocks, constituting neighborhood commercial zones. In Figure 11, as we move away from the CBD we see almost entirely tan blocks, allowing for a max building height of 40

stores, restaurants, and shops, but commuting to the CBD is still required to access most offices. Figure: 10 & 11

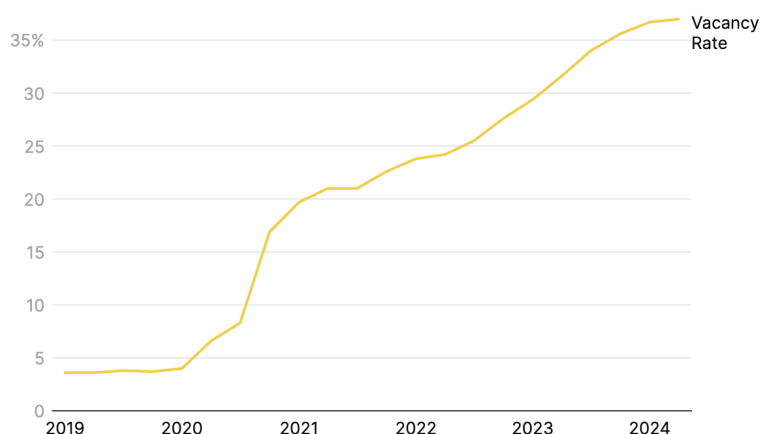
Section IV: Post COVID, Remote Work, and Flattening Bid Rent Curve in San Francisco

4.1 Post COVID Office Vacancy

In recent years San Francisco has solidified itself as the city experiencing the highest office vacancy rates in the United States. The record high vacancy rates are a product of post pandemic tepid attitudes towards returning to offices and the decentralization of work from the CBD. In mid 2019, just prior to the COVID outbreak, San Francisco experienced record low office vacancy rates sitting at less than 4% (D'Onfro 2020). Five years and one pandemic later office vacancy rates have skyrocketed to 37% in 2024 (D'Onfro 2024).

San Francisco's Record-Breaking Office Vacancy

After previously being one of the tightest commercial real estate markets in the country, San Francisco's office market has more recently been defined by record-breaking vacancy rates that peaked at 37% in the second quarter of 2024.



*Q2 2024 NUMBERS ARE PRELIMINARY
CHART: KEVIN TRUONG • SOURCE: CBRE

Source: CBRE

The main driver of office vacancy in San Francisco is the adoption of remote and hybrid work. Before the pandemic only 6% of workers in San Francisco were remote. A study from 2022 by the Bay Area Council Economic Institute found remote eligible jobs

now account for more than half of San Francisco's job market, a huge shift towards a decentralized remote office (BACEI). This increase in remote eligible jobs is largely due to San Francisco's high centralization of remote-friendly jobs in the tech and artificial intelligence industry. Roles such as software engineers, programmers, and data analysts are cloud based and unrequiring of face to face interaction, making proximity to the center less valuable to firms and employees. A 2022 study from McKinsey states "when people have the chance to work flexibly, 87 percent of them take it." Working from home allows people to save on commuting costs, spend more time with their families, and even "eighty-three percent of employees we surveyed cite the ability to work more efficiently and productively as a primary benefit of working remotely" (McKinsey). As we learned in class, information offices cluster in the center for access to face to face interaction, so without a need for in person interaction it is evident why people would choose to work from home; employees tend to value working from home, and with so many remote opportunities available in San Francisco jobs it is apparent why office vacancy has reached an all time high for the city.

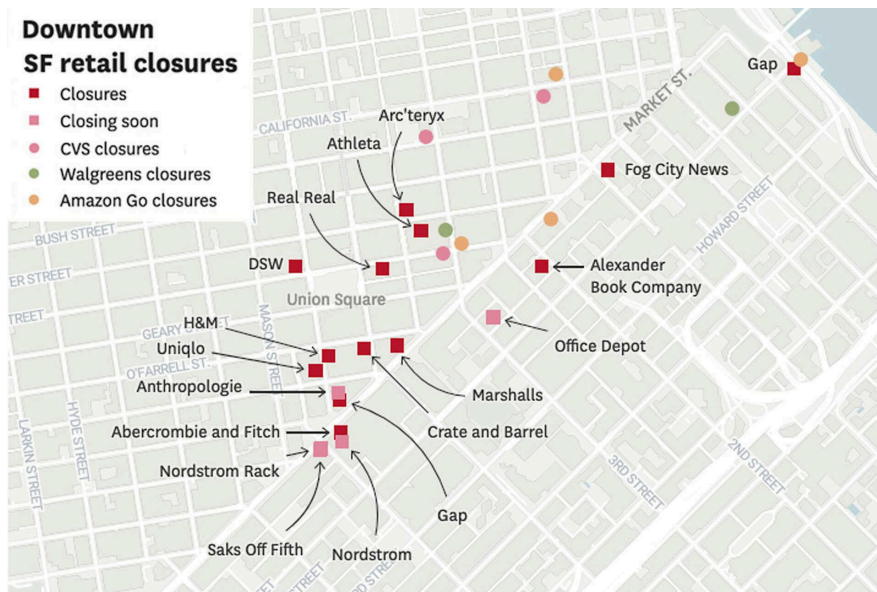
4.2 Firms Shift to Remote Work

The normalization of a shift to remote work has deteriorated demand and led San Francisco firms to minimize their office footprints. The sharp decrease in office space demand has caused tech industry giants to sublease, back out of leases, and move fully remote in order to prioritize cost efficiency and flexibility. For example, in 2023 Meta backed out of its downtown lease at its 181 Fremont Street office before ever moving in (SFStandard), and Dropbox paid landlords \$79 million dollars in order to get rid of more than a quarter of its SF headquarters in an effort to make remote work its primary model (Business Insider). This move by dropbox aimed to decrease costs, is in addition to them already subleasing 750,000 square feet of their office space. In 2022 Yelp fully closed down its downtown San Francisco headquarters with its CEO Jeremy Stoppelman pushing for a fully remote business operation. Major San Francisco tech firms are vacating the downtown district in order to cut their losses on high cost unused downtown offices and adapt to a workforce that no longer requires close proximity interactions. This exodus has led to a "ghost town" narrative for San Francisco's

downtown district. San Francisco's booming pre-COVID business district is now riddled with vacant buildings, crime, drugs, and homelessness making it an undesirable place for firms, stores, and households to locate.

4.3 San Francisco's Undesirable Downtown District

The mass office vacancies have not just affected the tech and financial sectors, but they have created a widespread ripple effect on stores, restaurants, and services. With fewer workers downtown San Francisco has seen a sharp decline of foot traffic in its business centers leaving hundreds of stores without a consistent customer base. As a result, many cafes, corner stores, restaurants and shops have had to close down due to a lack of sales. According to the San Francisco Standard, "Approximately 203 retailers were open on the streets surrounding Union Square in 2019. By May 2023, only 107, or 53%, were still in business." As an SF native, walking around downtown you see numerous storefronts boarded up and glass windows broken and shattered, making it seem like an abandoned, undesirable rural area.



Source: SF Chronicle

Not only do these store closures impose a negative public perception on downtown that deters locals and tourists, but it massively impacts San Francisco public

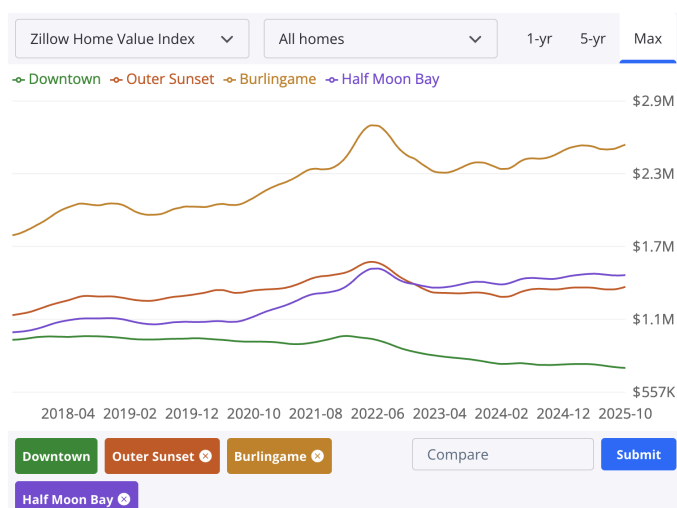
funding. The city of San Francisco collects taxes from sales and property that help fund the city's public services. Without commercial activity, the city loses millions of dollars in tax revenue that help fund police, street cleaning, homeless organisations, and public transit. The loss in tax revenue, said to be around "\$484 million dollars" by the Office of the Treasurer and Tax Collector, has led to less police downtown, more homeless on the streets, and a fall in public transit operations. As public services decrease and police activity diminishes, street conditions worsen, augmenting theft and violent crimes (US Chamber of Commerce). Disorder and unsafe conditions post-COVID deter residents from locating near downtown, further reinforcing the migration of firms and residents away from San Francisco's central business district.

4.4 San Francisco's Flattening Bid Rent Curve

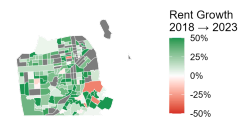
The combination of office vacancies, remote work adoption, retail closure, and increased crime have significantly flattened San Francisco's bid rent curve with hundreds of thousands of locals moving from the city center to more suburban areas of San Francisco. As mentioned previously, the majority of tech and professional jobs in San Francisco have transitioned to work from home agendas with around 40% of offices vacant (SF Gov). This digital shift has decoupled employment from physical proximity to Downtown, reducing the premium that workers and companies once placed on central locations. With work decentralized, many workers have moved outward for more space, affordability, and safety. Instead of clustering near the central city, professionals have dispersed to the East Bay, North Bay, South Bay, or even out of the region leaving "pricey housing markets like San Francisco or New York in favor of new hometowns, sometimes called "Zoom towns." (NYT). Furthermore, those who "have to go to the office only two or three days a week, moved and accepted lengthier "super commutes" in exchange for cheaper housing and more space" (NYT). This migration to cheaper regions underlines a core concept we've studied in class: the tradeoff between commute time and housing prices. Traditionally, people paid a premium to live closer to the urban core to reduce commuting costs and access face to face interaction, resulting in higher land values closer to the CBD. But remote and hybrid work have weakened that tradeoff. The Bay Area as a whole lost about 3.2% of its population, about 250,000

people, between 2020 and 2022 (United States Census Bureau). Workers pay a higher commuting cost a few times a week in exchange for larger homes, lower rents, and better quality of life. “San Francisco saw the nation’s biggest jump in the percentage of the workers living 50 or more miles from their jobs from 2012 to 2022...By 2022, 18.7% of San Francisco’s workforce lived 50 or more miles away from their jobs, up from 13.7% a decade earlier” (SF Chronicle). As a result, the demand curve has redistributed, flattening the bid-rent curve that once made central locations significantly more expensive.

The shift in firms and residential location choice has augmented demand and prices in outer areas, depressing demand and rent in SF’s central neighborhoods such as SoMa and the financial district. Digitalization of the workforce has caused firms to no longer have to cluster close together to exchange information, diminishing the value of proximity in the center. Where housing and commercial rents pre-COVID used to decline with distance from the urban core, that gradient has flattened. Suburban rents have risen and city-center rents have dropped flattening the San Francisco bid rent curve. This can be seen in some of San Francisco’s downtown housing prices dropping “by nearly 30 percent since 2019”, indicating a softening in real estate demand in the city center (newsweek). On the other hand we can see an increase in price of San Francisco’s non central residential districts such as the Outer Sunset, Burlingame, and Half Moon Bay. The graphs below shows how housing prices have changed since COVID 19.



Change in Median Gross Rent (2018–2023)
Census Tracts — San Francisco County, ACS 5-Year Estimates



Source: U.S. Census Bureau, ACS 5-Year (B25064_001)

Source: Zillow

As SF's downtown prices are steadily dropping, neighboring suburban areas are increasing, experiencing their peak in 2022 of the COVID pandemic when all work moved remote. The 2022 peak in outer SF housing prices directly underlines how remote work untethered people from downtown, triggering a wave of migration to cheaper, more livable areas. If remote work continues to become the standard for SF firms, with more firms going fully remote, we will see the bid rent curve continue to flatten with prices and demand in outer regions continuing to rise, as the CBD demand drastically falls. The graph illustrates a flatter bid rent curve with demand and prices bolstered as people move further away from San Francisco's city center. Although rents have begun to partially rebound in some high-demand downtown areas by 2025 with an influx of new AI firms, the new normal bid rent curve still reflects a more dispersed geography of housing demand. Proximity to downtown remains valuable but is no longer essential as hybrid work continues to reshape San Francisco's urban economy.

Section V: Conclusions

Modern land rent trends in San Francisco no longer align with traditional monocentric urban area patterns discussed in class. The financial district still remains a central hub for commercial office firms, with a majority of its workforce commuting from suburbs throughout the county. However, COVID has flattened the bid rent curve as an increasing number of workers disperse to further from the center for remote work and avoiding densely populated areas. While surrounding neighborhoods are densely populated, disamenities associated with dense urban centers and commercial zoning have resulted in urban decay within the heart of the CBD, where a lack of available amenities has lowered residential rent prices and demand in the area. Land rent prices in the suburbs have continued to grow post pandemic, as CBD accessibility improves through public transport and homeowners increasingly prefer larger lot sizes.

The findings show that San Francisco's bid-rent curve hasn't disappeared, but has been redefined. Accessibility, and face-to-face time still matters, but through digital connectivity, historic locational advantages have dwindled. As long as remote or hybrid

work continues to be a key feature within the region's economy, San Francisco's urban form and land-rent gradients will continue to shift away from the classic monocentric city model that had defined the city for over a hundred years.

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